

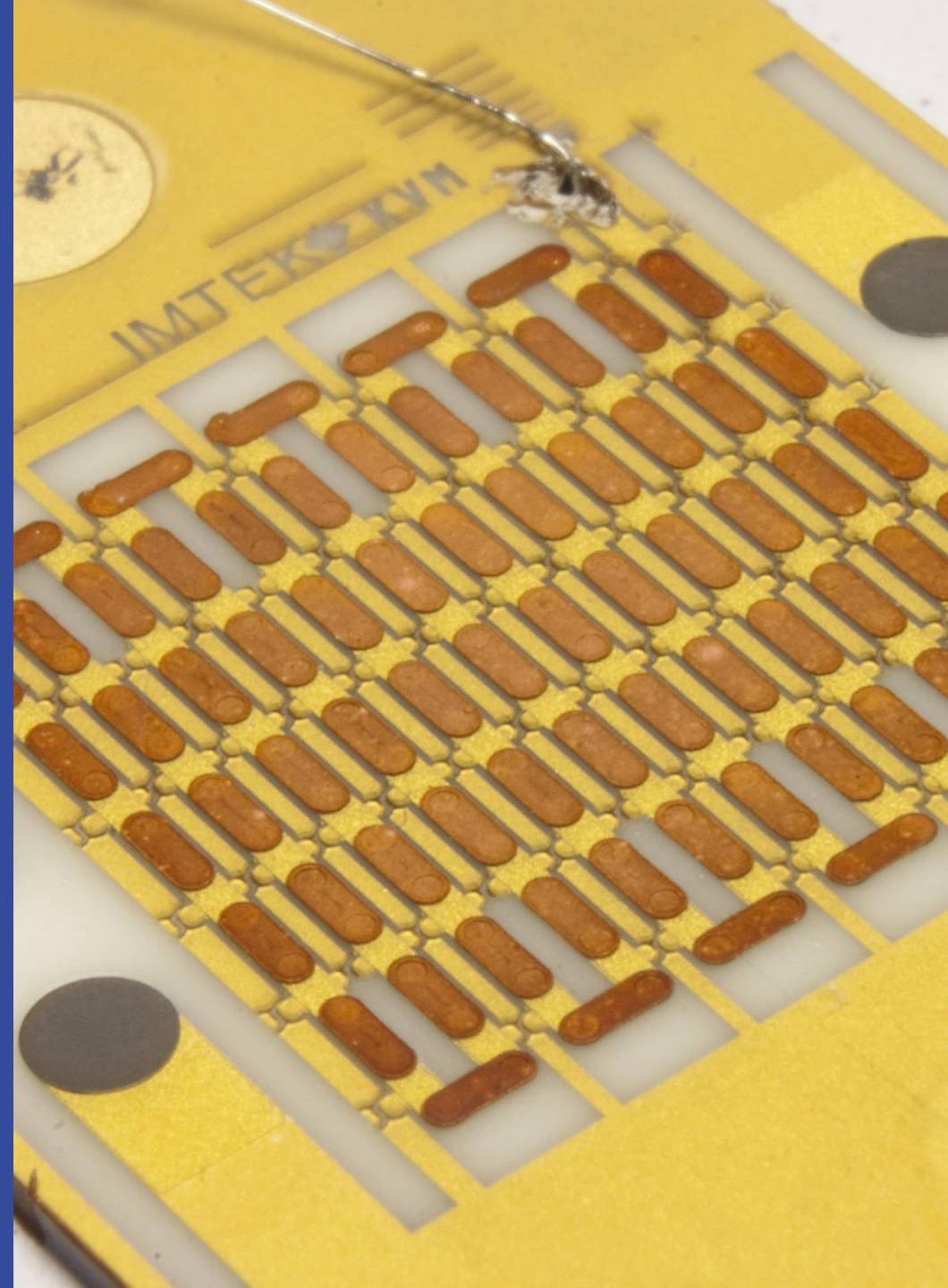
Micro Energy Harvesting

Chapter 7: Storage Concepts I - Capacitive

Micro Energy Harvesting SoSe 2026

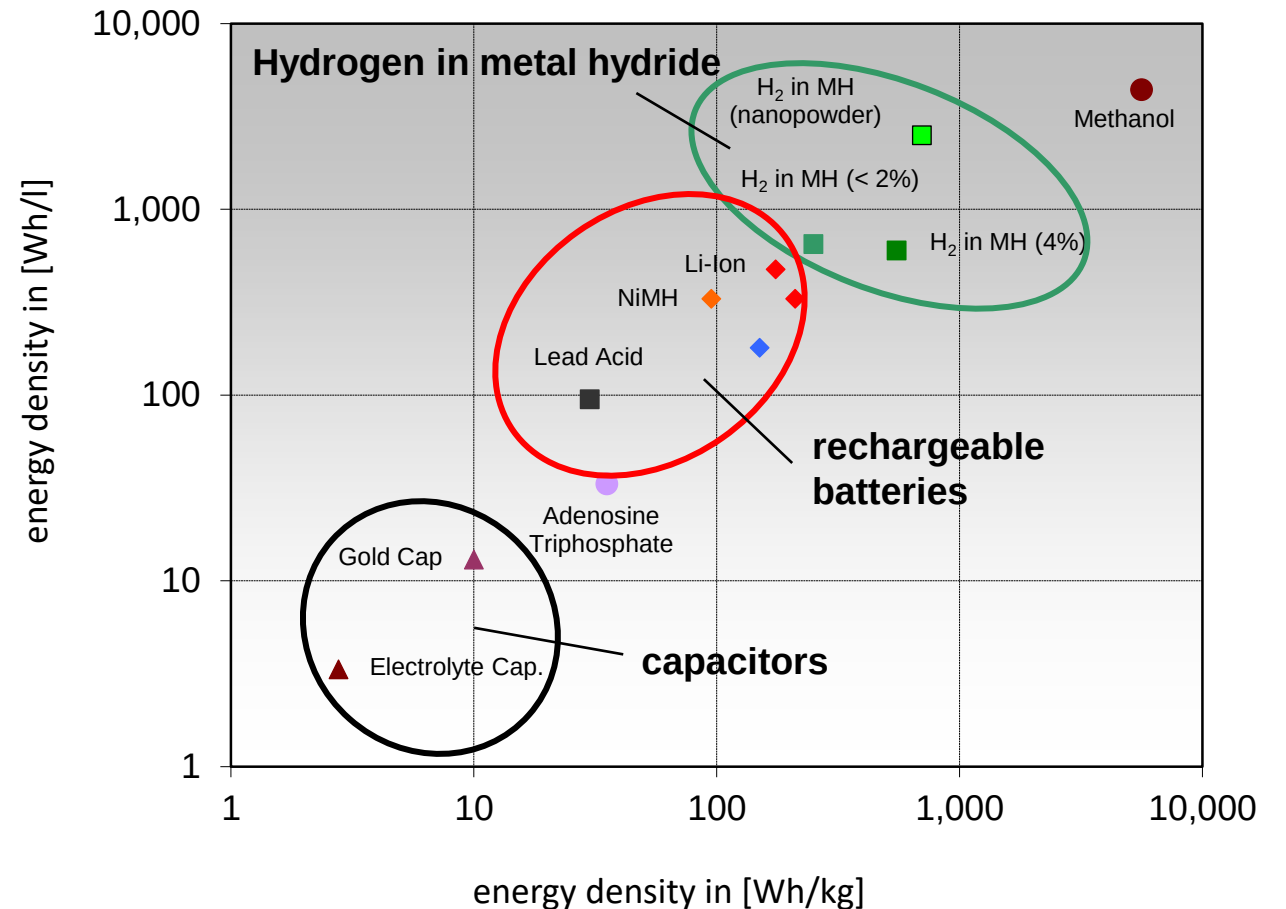
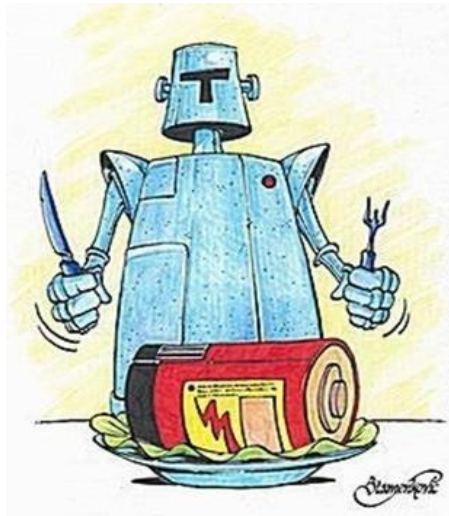
IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank Goldschmidtboing



Energy Storage for Embedded Systems

- today mostly electric
- slowly emerging alternatives (e.g. thermal)
- design conflict between ...
 - storage capacity,
 - size/weight and
 - lifetime
 - ...



Refs: J. Brodd et al, *J. Electrochem. Soc.*, 151 (3), 2004, K1-K11 and HERA Hydrogen Storage Solutions, D

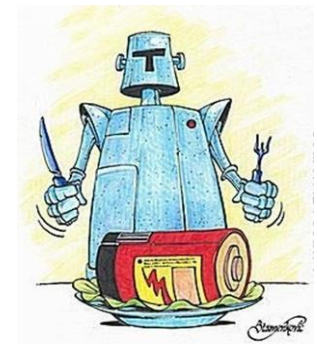
Energy Storage for Embedded Systems Requirements...

... by energy-autonomous embedded systems

- highly variable energy input from the generator
- short to long periods of zero harvested power
- Long operational lifetime required (up to 10s of years)
- highly variable energy demand from the embedded system
 - sleep mode: *ultra low power*
 - sensing and signal processing: *low power*
 - wireless communication: *high power*

... for energy-autonomous embedded systems

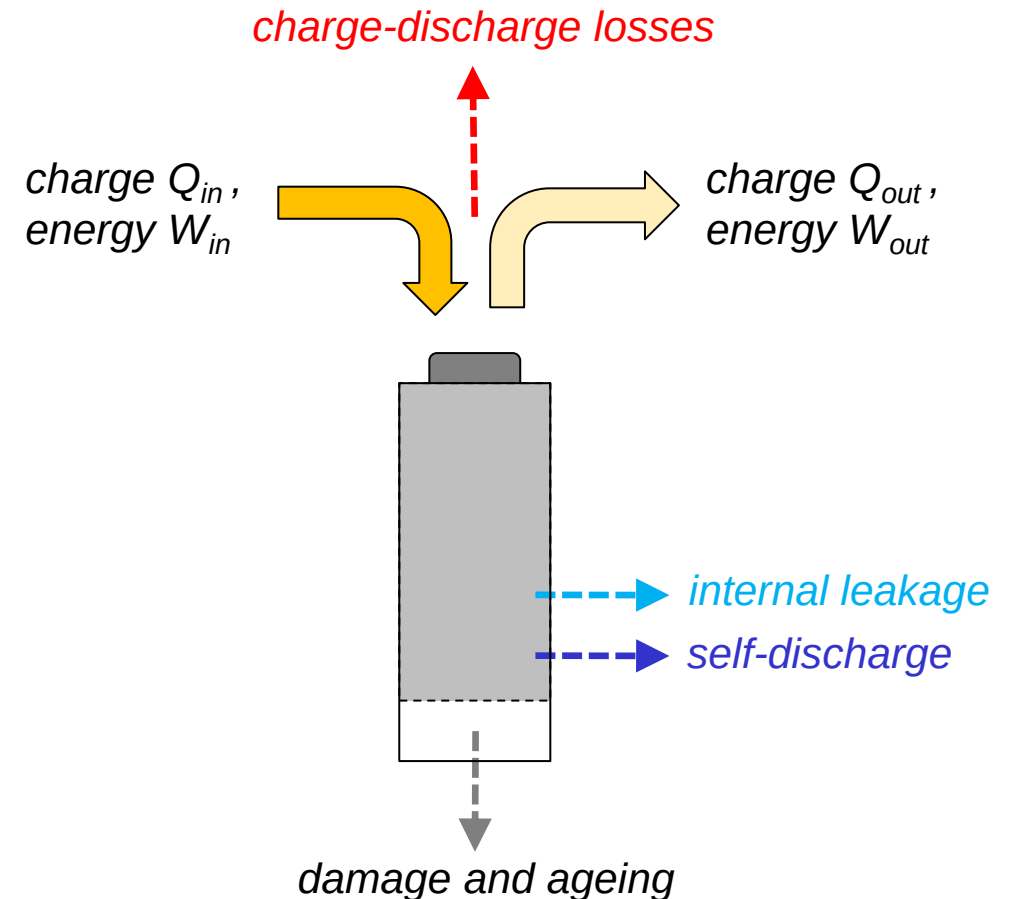
- high charge-discharge and energy efficiency
- **appropriate** size, weight and capacity
- appropriate cycle number for charge/discharge
- tolerance against low **and** high charging/discharging currents
- small self-discharge and small ageing effects, long lifetime
- others (e.g. temperature tolerance)



Electrical Energy Storage

Efficiencies and Losses

1. Efficiencies of the charge-discharge process
 - Coulomb efficiency
 - Energy efficiency
2. Continuous internal losses
 - leakage current
3. Self-discharge
 - self-discharge rate SDR
4. Damage and ageing effects: loss of capacity due to ...
 - mechanical stresses and internal material dislocations
 - memory effects
 - leakage, dry-out of electrolyte
 - over- or undertemperature
 - overcharge or deep discharge ,...



Electrical Energy Storage

Charge-discharge Efficiency or „Coulomb Efficiency“ η_C

- based on the **electric charge** in [As] that is transferred over the system
- defined as ratio of delivered charge Q_{out} vs. input charge Q_{in} during one charge-discharge cycle
- no representation of voltages during charge and discharge
- depends on ...
 - type of storage (capacitor, battery,...)
 - operation conditions (currents, temperatures, ...)

$$\eta_C = \frac{Q_{out}}{Q_{in}}$$

Example: Charging and discharging of a 1000 mAh NiCd battery from a 1.5 V power supply

- standard charging: 14 hrs at 100 mA and 1.5 V
- standard discharging: 10 hrs at 100 mA and 1.2 V (average voltage)

$$\left. \begin{array}{l} Q_{out} = 100 \text{ mA} \cdot 10 \cdot 3600 \text{ s} \\ Q_{in} = 100 \text{ mA} \cdot 14 \cdot 3600 \text{ s} \end{array} \right\} \Rightarrow \eta_C = \frac{3600 \text{ As}}{5040 \text{ As}} = 71.4 \%$$

Electrical Energy Storage

Energy Efficiency η_E

- based on the **electric energy** in [J] or [Ws] that is stored in the system
- defined as ratio of delivered energy E_{out} vs. input energy E_{in} during one charge-discharge cycle
- takes voltages and currents into account
- depends on ...
 - type of storage (capacitor, battery,...)
 - operation conditions (currents, temperatures, ...)

$$\eta_E = \frac{E_{out}}{E_{in}}$$

Example: Charging and discharging of a 1000 mAh NiCd battery from a 1.5 V power supply

- standard charging: 14 hrs at 100 mA and 1.5 V
- standard discharging: 10 hrs at 100 mA and 1.2 V (average voltage)

$$\left. \begin{array}{l} E_{out} \approx 1.2 \text{ V} \cdot 100 \text{ mA} \cdot 10 \cdot 3600 \text{ s} \\ E_{in} = 1.5 \text{ V} \cdot 100 \text{ mA} \cdot 14 \cdot 3600 \text{ s} \end{array} \right\} \Rightarrow \eta_E \approx \frac{4320 \text{ Ws}}{7560 \text{ Ws}} = 57.1 \%$$

Electrical Energy Storage

Self-discharge Rate SDR

- mostly relevant - and in use - for **rechargeable batteries**
- defined as relative loss of stored charge C in [As] over a certain period of time
- generated by ...
 - internal leakage currents
 - various electrochemical side effects
 - external influences (e.g. temperature, mechanical stress)

$$SDR = \frac{\Delta C}{C_0 \cdot \Delta t}$$

C_0 : *initial capacity* [As]
 ΔC : *loss of capacity over* Δt [As]
 Δt : *test interval* [s]

Electrical Energy Storage

Leakage Current

- mostly relevant - and in use - for **capacitors**
- defined as leakage current I_{leak} in [A] due to internal losses
- depends on ...
 - applied or actual voltage
 - time (short-term vs. long-term leakage effects)
 - external influences (e.g. temperature)
- standard measurement protocols available, but also ...
- different proprietary definitions by the suppliers of capacitors

Energy Storage for Embedded Systems

Global Properties

Capacitors

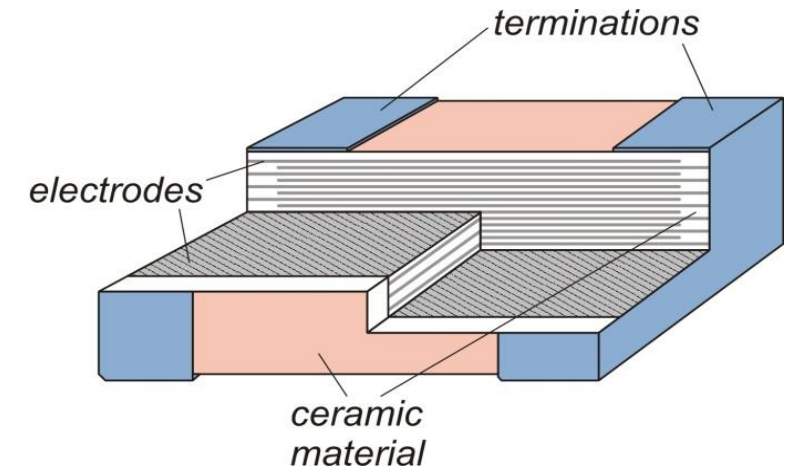
- dielectric capacitors
 - dielectric multilayer capacitors (MLCC)
 - Tantal capacitors
 - electrolyte capacitors
 - Gold caps ...
- **charge storage**
 - (mostly) linear voltage-charge characteristics: $Q = C \cdot U$
 - dischargeable to 0 V
 - internal leakage currents ➡ electrical self-discharge
 - minimum charge current to overcome leakage effects
 - requires an overvoltage protection

Rechargeable batteries

- Pb
 - NiCd, NiMH
 - Li-Ion
 - Li-Po
- **chemical energy storage**
 - non-linear voltage-charge characteristics
 - usually not dischargeable to 0 V
 - chemical and electrical self-discharge
 - requires a dedicated charge and discharge management

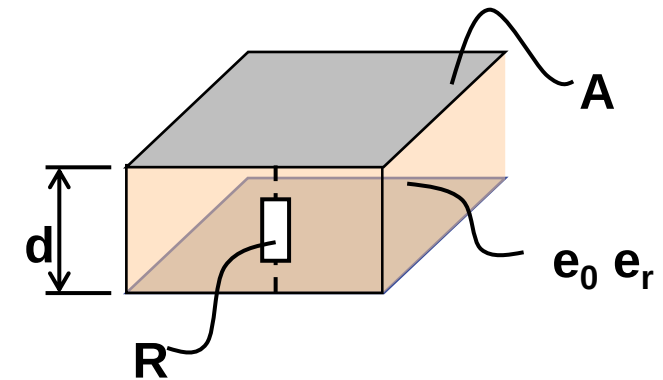
Multilayer Ceramic Capacitors (MLCCs) Design

- Multilayers of high dielectric ceramic material and metal layers
- Various classes of ceramic material in use
- Also: “multi layer chip capacitors”



$$C = \epsilon_0 \cdot \epsilon_r \cdot \frac{A}{d}$$

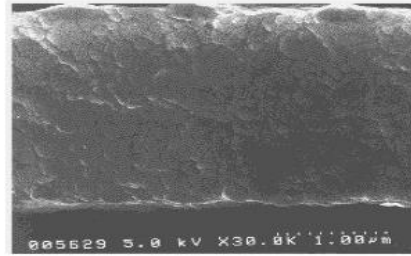
$$R_P = \rho \cdot \frac{d}{A}$$



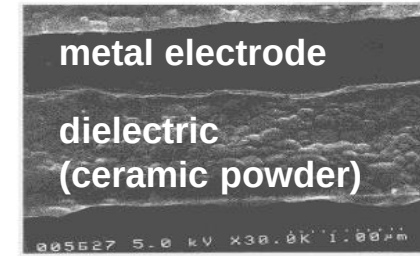
Multilayer Ceramic Capacitors (MLCCs)

Trends

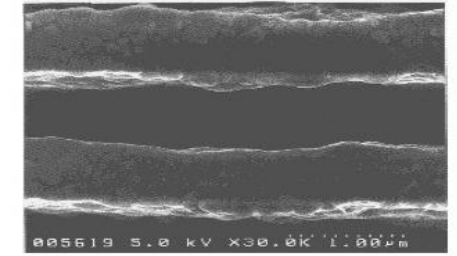
- Decreased layer thickness and increased No. of layers
→ higher capacity
- Size reduction (smart devices)



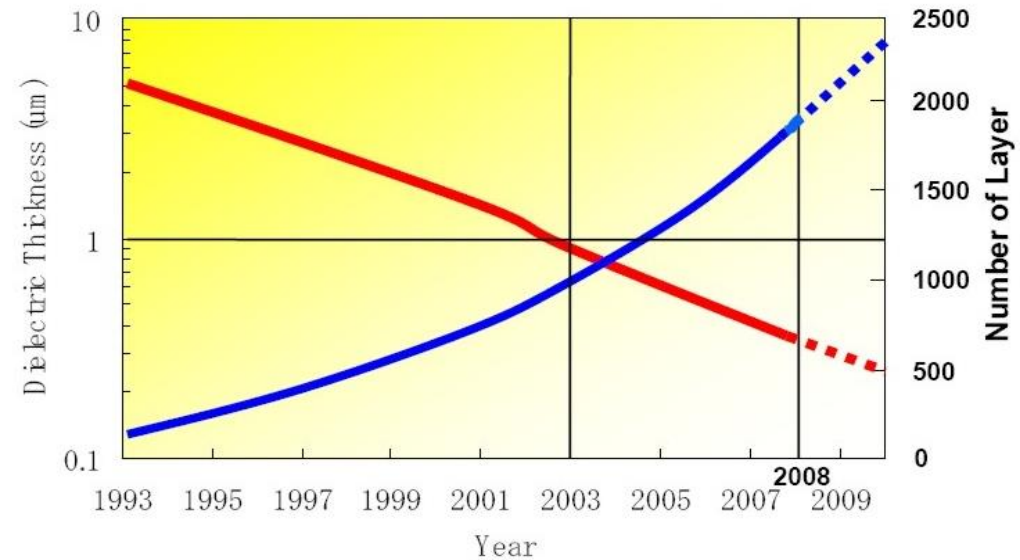
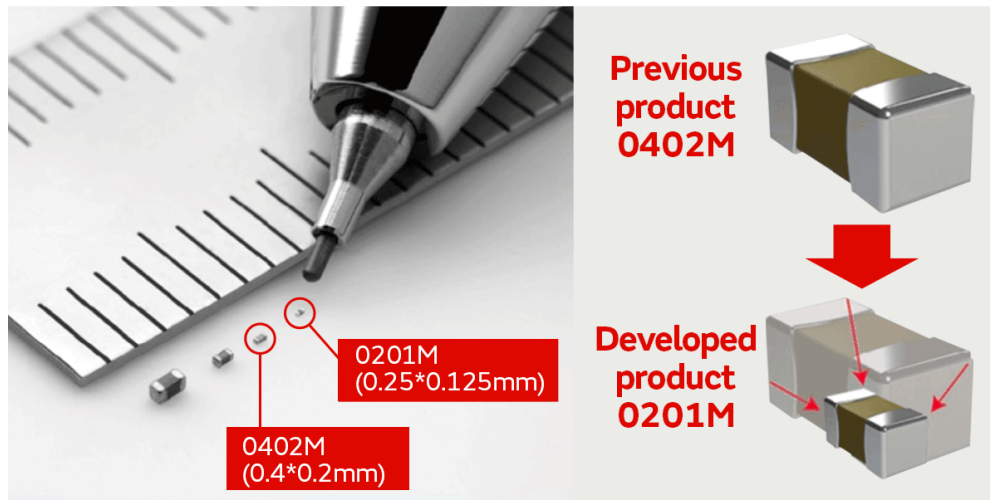
2000: $d = 2 \mu\text{m}$



2003: $d = 1 \mu\text{m}$



2007/2008: $d = 0.5 \mu\text{m}$



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Multilayer Ceramic Capacitors (MLCCs)

Electrical Equivalent Circuit

C nominal capacitance

may depend on the applied voltage

R_p parallel resistance, mostly of the dielectric material

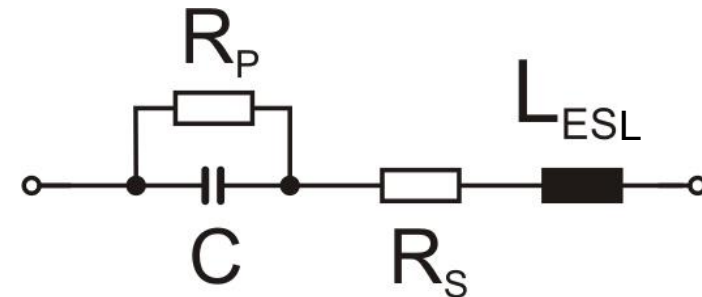
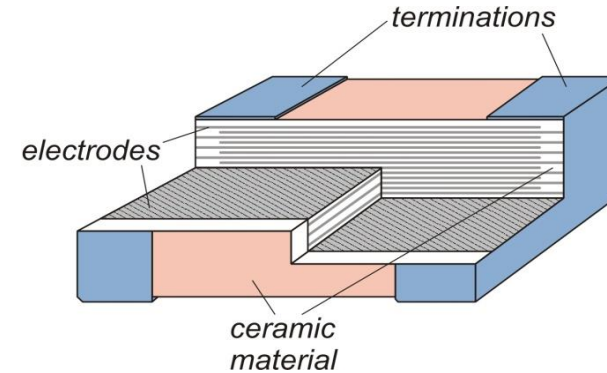
depends on material behavior, rated capacitance and geometric sizes

R_s serial resistance

given by the electrode and termination resistances

L_{ESL} Equivalent Serial Inductance ESL

given by the electrode and termination inductances

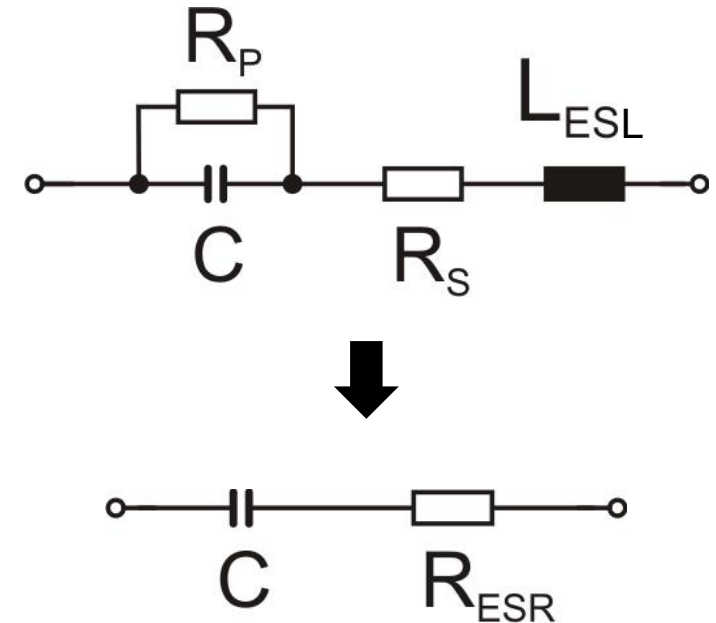


Multilayer Ceramic Capacitors (MLCCs)

Simplified Electrical Equivalent Circuit

- used for the calculation of internal losses under AC conditions
- inductive elements L_{ESL} are frequently omitted
- R_P and R_S combined into one element R_{ESR}
- R_{ESR} given via the loss factor $\tan(\delta)$ in the datasheet
- R_{ESR} **E**quivalent **S**eries **R**esistance ESR
*given by the electrode and termination resistances
and losses in the dielectric material*

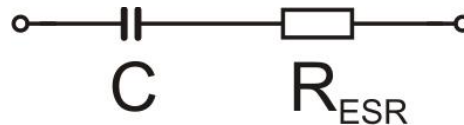
$$\text{for } R_S \ll R_P : R_{ESR} \approx \frac{R_P \cdot R_S}{R_P + R_S}$$



Multilayer Ceramic Capacitors (MLCCs)

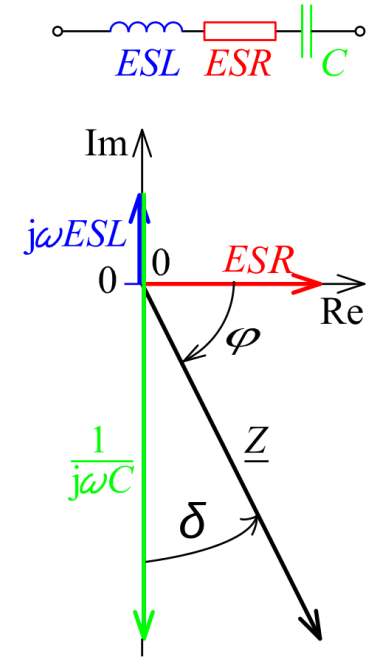
Loss factor $\tan(\delta)$

- Defines the ohmic losses in the internal resistances R_{ESR} of a capacitor
- An angle for the representation of ohmic losses?
 - ➔ R_{ESR} cannot be measured directly, however it produces a phase shift angle d from the 90° value found between AC current and AC voltage in an ideal capacitor
 - ➔ easy to be measured via the $\tan(\delta)$
- usually given for a certain range of frequencies or a single frequency in the datasheet
- influenced by summarized dielectric and resistive losses of the dielectric material



ideal capacitor: $Z_C = \frac{1}{j\omega C}$

real capacitor: $Z = R_{ESR} + \frac{1}{j\omega C}$



Source: wikipedia.org

$$\text{loss factor: } \tan(\delta) = \frac{\text{Re}|Z|}{\text{Im}|Z|} = R_{ESR} \cdot \omega C$$

Multilayer Ceramic Capacitors (MLCCs)

Ceramic Dielectric Material

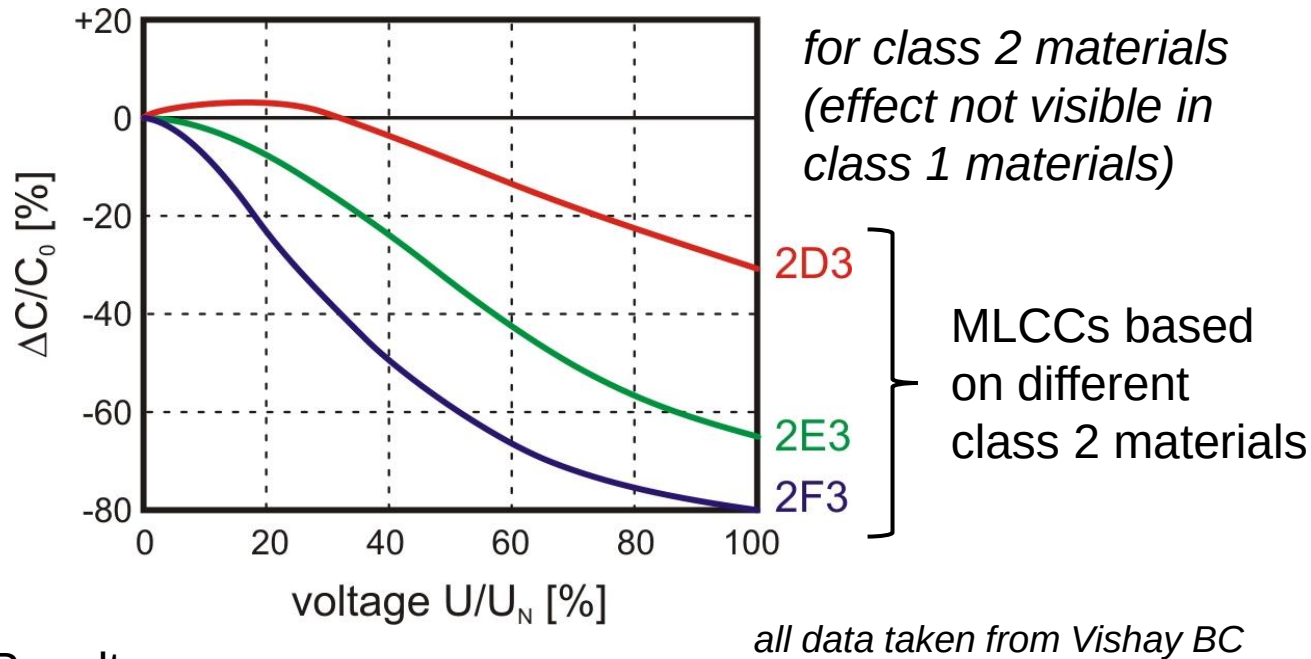
Two predominant material classes according to global properties:

- *Class 1 (low-K capacitors)*: mostly titanium dioxide or magnesium silicate
- *Class 2 (high-K capacitors)*: mostly alkaline titanates

	Class 1	Class 2
selected material codes	P100,NP0,N150,N750,N1500,SL0,S3N	X5R,X7R,Y5P,Z5U,Z5V,Y5V,Y5U
<i>typical characteristics of the electrical capacitance</i>		
stability	high	acceptable to low
dissipation factor $\tan(\delta)$	up to 1.5×10^{-3}	up to 35×10^{-3}
temperature dependence	adjustable (positive, negative, zero)	non-linear
voltage dependence	small	non-linear
insulation resistance R_p	$> 10 \text{ GOhm}$	$> 1 \text{ GOhm}$
degree of miniaturization	acceptable to high	very high

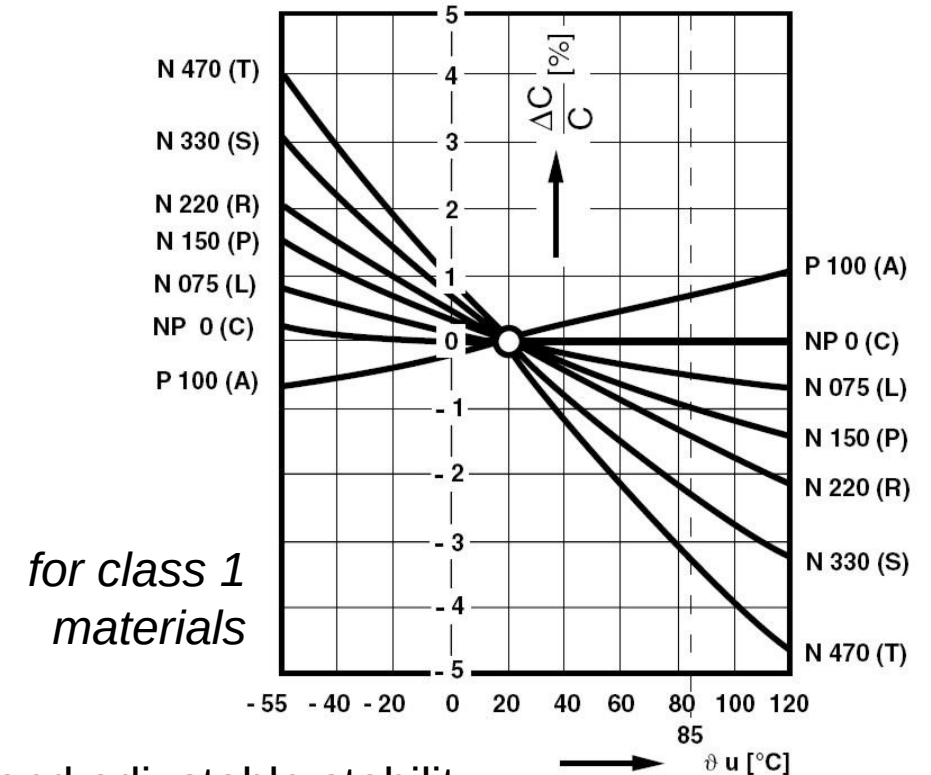
Multilayer Ceramic Capacitors (MLCCs)

Voltage and Temperature Dependence



Result

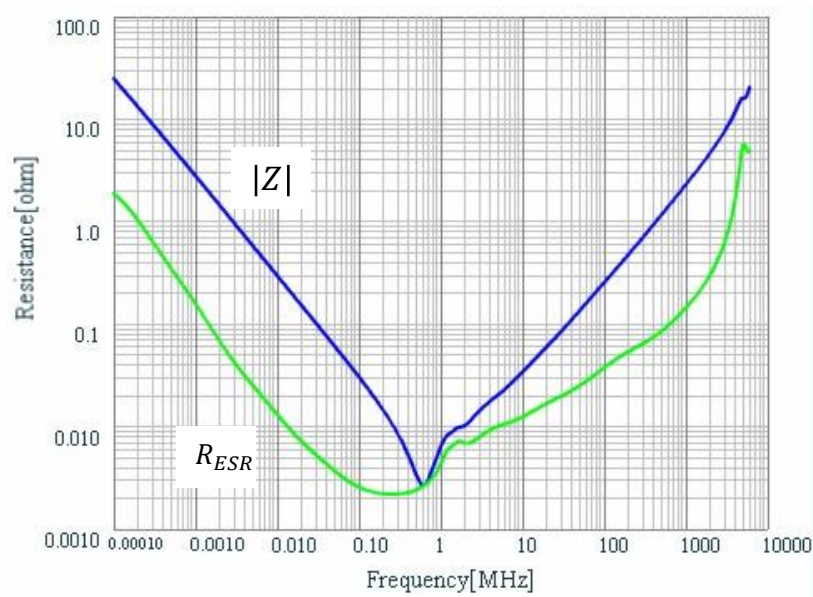
- class 1 materials: low energy density (due to lower ϵ_R) but high and adjustable stability
- class 2 materials: high energy density (due to higher ϵ_R) but low stability



Multilayer Ceramic Capacitors (MLCCs)

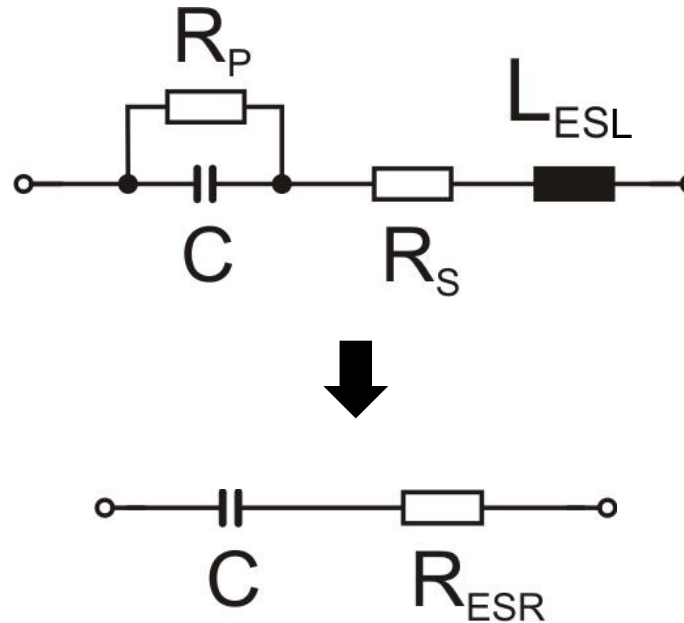
Frequency-dependence of R_{ESR} and Z

Murata GRM31CR60J107ME39



$$\text{impedance: } Z = R_{ESR} + \frac{1}{j\omega C}$$

$$\text{loss factor: } \tan(\delta) = R_{ESR} \cdot \omega C$$



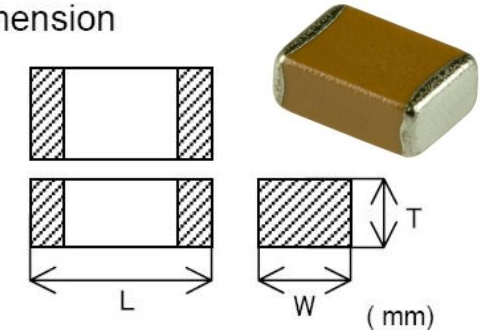
class 2: $C = f(U) \rightarrow$

$\tan(\delta) \rightarrow$

$R_p \rightarrow$

1206 X5R 100uF 6.3V

1. Dimension



L	W	T
3.2 +/-0.3	1.6 +/-0.3	1.6 +/-0.3

2. Cap,DF,IR

Capacitance,DF 120Hz , 0.5Vrms

IR 6.3V , 60s

Item	Spec
Cap.[uF]	80 to 100
DF	0.15max
IR[M ohm]	1.0 min

Multilayer Ceramic Capacitors (MLCCs)

Frequency-dependence of R_{ESR} and Z

Calculation vs. measurement of R_{ESR}

Murata GRM31CR60J107ME39

values f [Hz]	measured tan(δ)	measured C [μ F]	datasheet R_{ESR} [Ω]	calculated R_{ESR} [Ω]
0,1	0,07	65		1713,976
1	0,07	65		171,398
10	0,07	65		17,140
100	0,0747	63,7	1,913	1,866
200	0,0796	61,1	1,035	1,037
1000	0,0547	56,5	0,156	0,154
10000	0,0447	53	0,014	0,013
100000	0,0831	52	0,0026	0,003

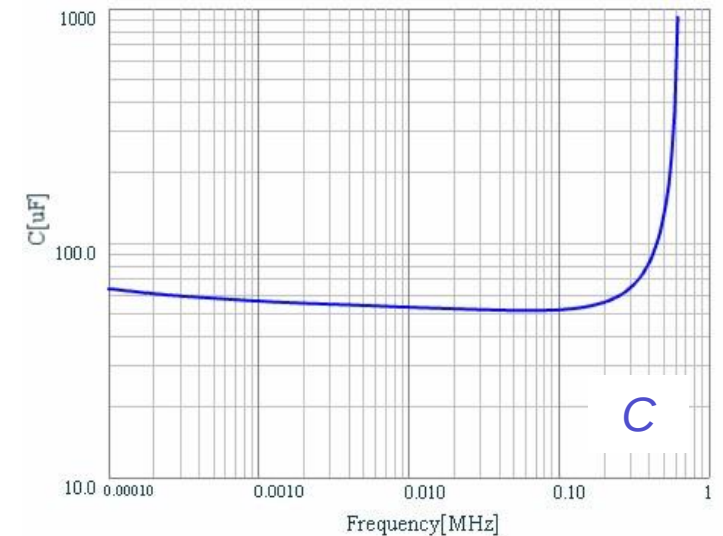
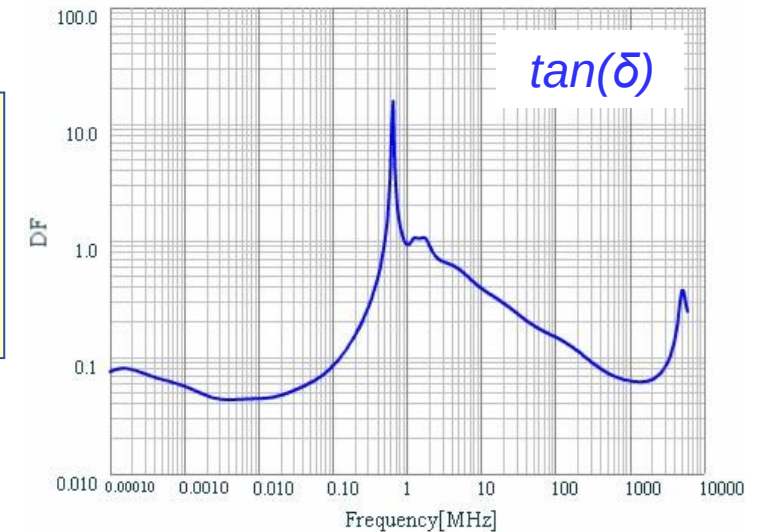
Result

in green: assumed values

- increase of R_{ESR} with lower frequencies
 ➔ higher losses at lower frequencies
- datasheet values of R_{ESR} are obviously **calculated** from tan(δ) and C

$$|Z| = \sqrt{R_{ESR}^2 + \left(\frac{1}{\omega C}\right)^2}$$

$$\tan(\delta) = R_{ESR} \cdot \omega C$$

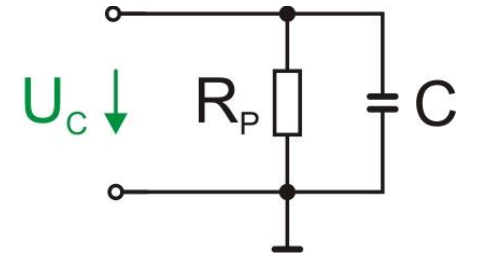


Multilayer Ceramic Capacitors (MLCCs)

Self-discharge and other Parameters

- **Self-discharge time** τ : time constant of the internal RC element

$$\tau = R_p \cdot C$$



- **50% energy drop time** t_{50} : time period for a discharge of 50% of the initially stored energy W_0

Discharge curve of a disconnected capacitor:

$$U_C(t) = U_0 \cdot \exp\left(-\frac{t}{\tau}\right); \quad W(t) = \underbrace{\frac{1}{2} C \cdot U_0^2}_{W_0} \cdot \exp\left(-\frac{2 \cdot t}{\tau}\right) \Rightarrow t_{50} = 0.35 \cdot \tau$$

- **Continuous power loss** P_{loss} : leakage through the parallel resistance R_p

$$P_{\text{loss}}(t) = \frac{U_C^2(t)}{R_p}$$

Multilayer Ceramic Capacitors (MLCCs)

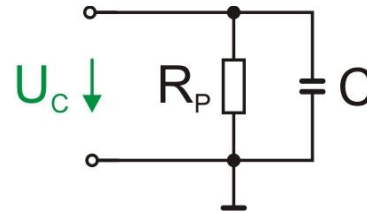
Insulation Resistance (IR or R_p)

- Example: Murata GRM series (*GRM43SR60J107ME20L*)
- Calculated R_p acc. to global datasheet:

$$C > 0.047 \mu F: R_p = \frac{500}{C[F]} \Omega$$

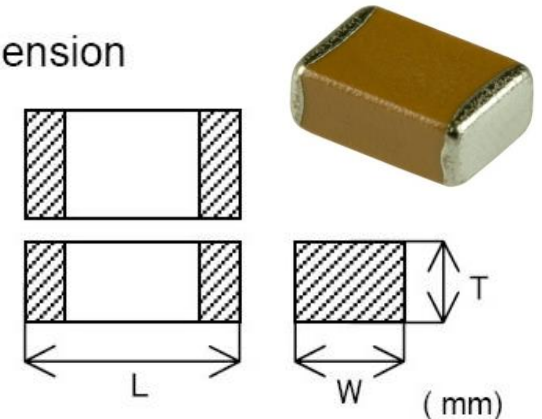
$$C = 100 \mu F: R_p = \frac{500 \Omega}{100 \cdot 10^{-6}[F]} = 5 M\Omega$$

- Minimal R_p according to detailed datasheet: $R_{p,min} = 1 M\Omega$



1206 X5R 100uF 6.3V

1. Dimension



L	W	T
3.2 +/-0.3	1.6 +/-0.3	1.6 +/-0.3

2. Cap,DF,IR

Capacitance,DF 120Hz , 0.5Vrms

IR 6.3V , 60s

Item	Spec
Cap.[uF]	80 to 100
DF	0.15max
IR[M ohm]	1.0 min

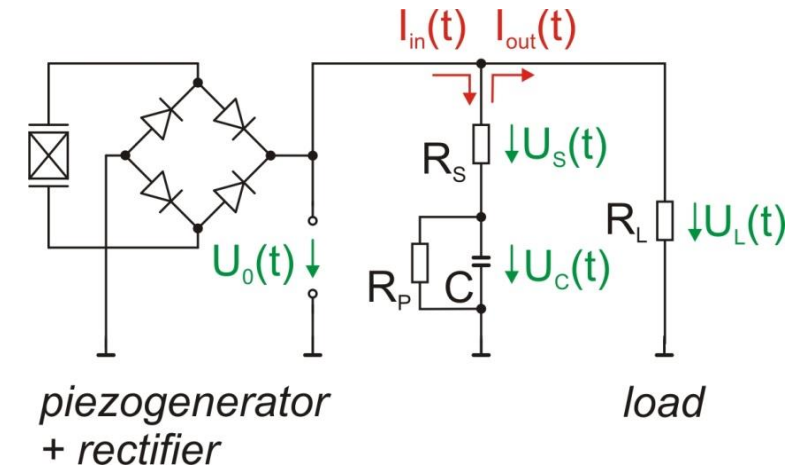
$R_{p,min} \rightarrow$

Multilayer Ceramic Capacitors (MLCCs)

Internal Losses During Charge and Discharge

Example: Murata GRM43SR60J107ME20L – 100 μF as buffer capacitor for an embedded system (R_L)

- generator: piezoelectric generator; delivers $U_{\max} = 2.66\text{V}$ at $f = 100\text{ Hz}$
- buffered system voltage: 2.3 V
- wireless communication: $220\text{ }\mu\text{W}$
- sensing and data processing: $15\text{ }\mu\text{W}$
- power management: $10\text{ }\mu\text{W}$
- standby power: $5\text{ }\mu\text{W}$



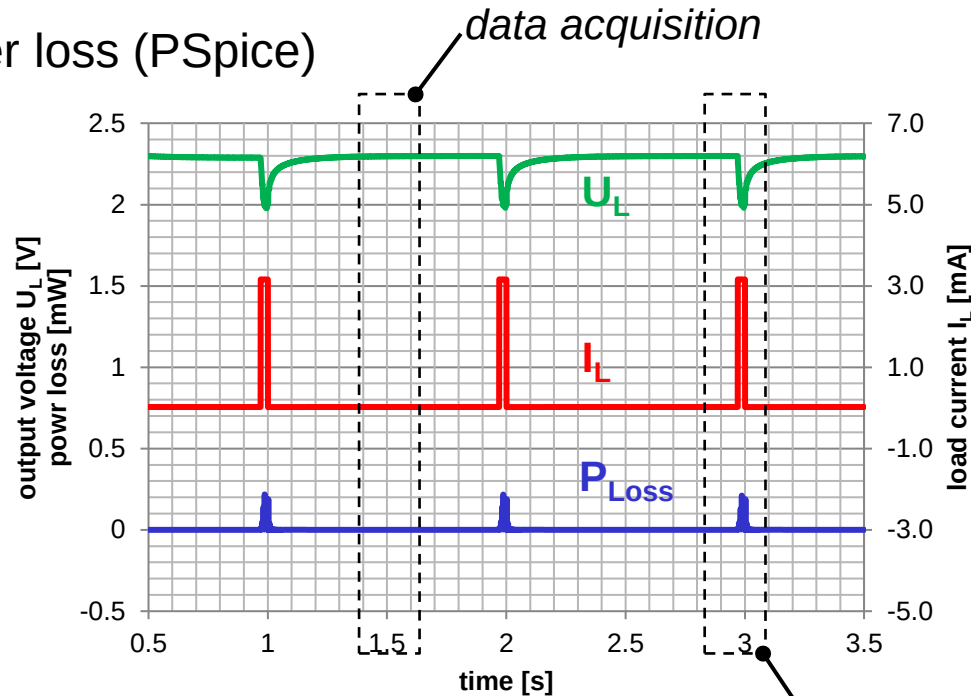
Observations

- time-variable voltage drop $U_S(t)$ over R_S during charge and discharge of the buffer capacitor
 - ➔ ripple voltage added to $U_L(t)$
 - ➔ power losses in R_S

Multilayer Ceramic Capacitors (MLCCs)

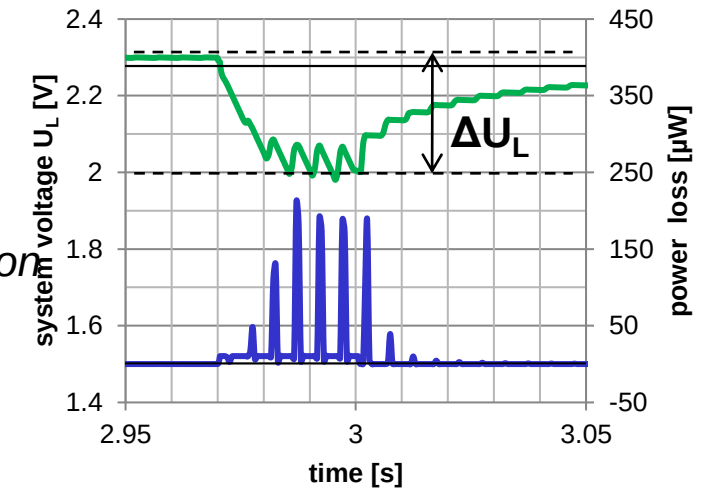
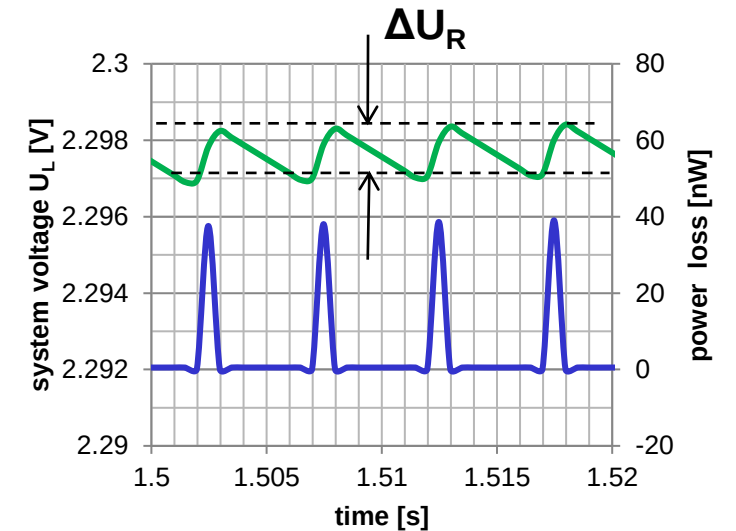
Internal Losses During Charge and Discharge

Ripple effect and power loss (PSpice)



Result

- ripple ΔU_R over the system voltage U_L
- high power loss in R_S and high voltage breakdown ΔU_L during periods of high load current



Multilayer Ceramic Capacitors (MLCCs)

Internal Leakage Currents and Self-discharge

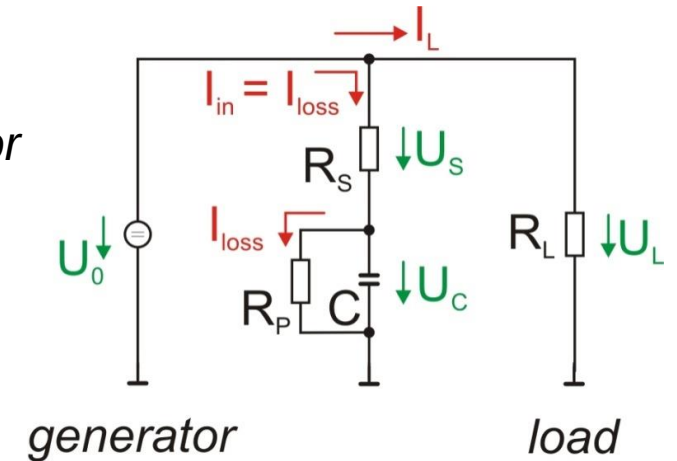
- *Example: Murata GRM43SR60J107ME20L – 100 μF /6,3 V as buffer capacitor*
 - *buffered system voltage: 2.3 V*
 - *wireless communication: 220 μW*
 - *sensing and data processing: 15 μW*
 - *power management: 10 μW*
 - *standby power: 5 μW*

$$\left. \begin{array}{l} U_C \approx U_0 = 2.3 \text{ V} \\ R_P = 5 \text{ M}\Omega \\ C = 100 \text{ }\mu\text{F} \end{array} \right\} \Rightarrow \begin{array}{l} P_{\text{loss}} = 1.06 \text{ }\mu\text{W}; \\ \tau = 500 \text{ s}; \\ t_{50} = 175 \text{ s}; \end{array}$$

Result

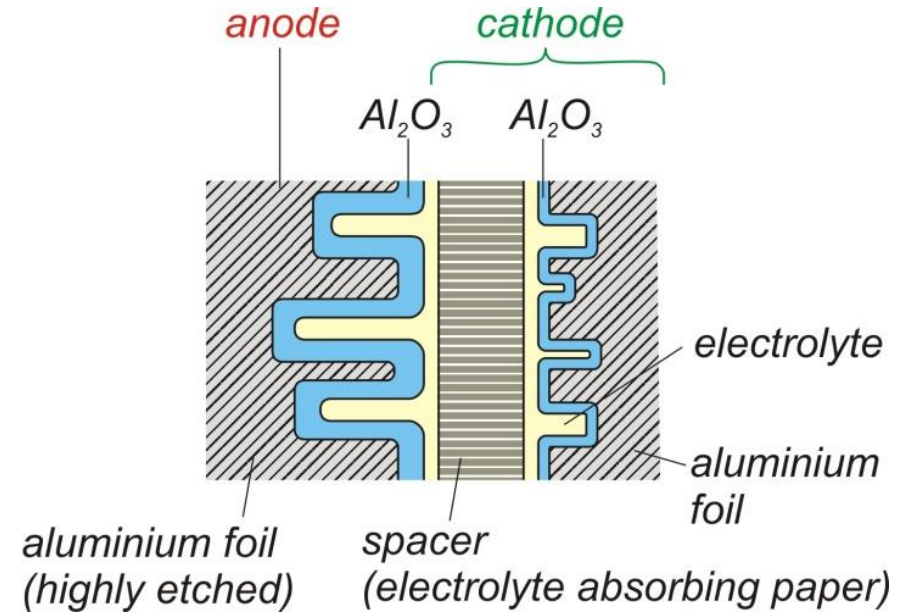
The relatively low parallel resistance R_P produces ...

- a high continuous additional power drain P_{loss} in operation (20% of the standby power)
- a rapid self-discharge within t_{50} (loss of 50% stored energy within 3 minutes)



Electrolytic Capacitors

Design and Internal Layout



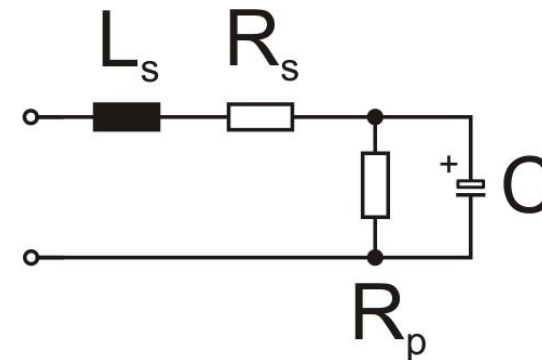
Electrical equivalent circuit

R_p : parallel resistance

R_s : equivalent series resistance („ESR“)

L_s : serial inductance

C : capacitance



Electrolytic Capacitors

Design and Internal Layout

Origin of leakage currents

- imperfect Al_2O_3 insulators (small effect)
- build-up of the oxide layers (temporary effect)
- defects in the Al_2O_3 insulators (temporary, self-healing)
- bypass currents
- ageing, ...

Equivalent circuit for leakage effects

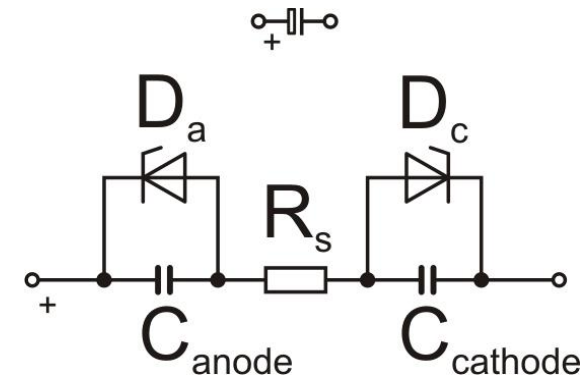
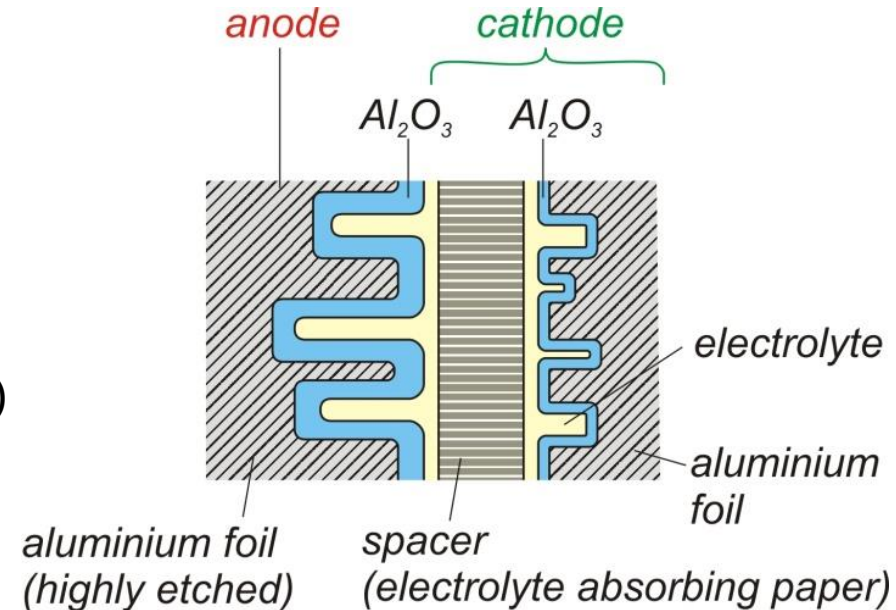
D_a : zener diode for anode breakthrough

D_c : zener diode for cathode breakthrough

R_s : equivalent series resistance („ESR“)

C_a : capacitance at the anode

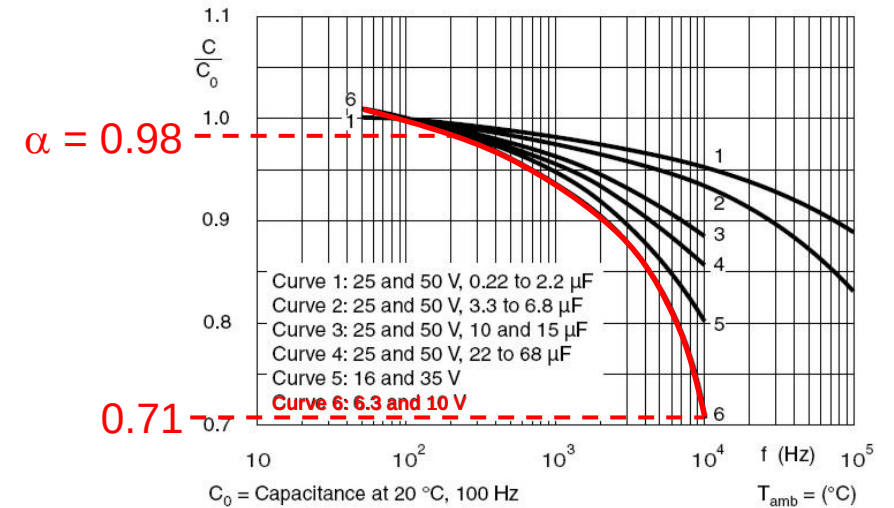
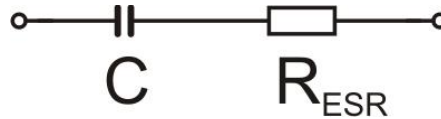
C_c : capacitance at the cathode



Electrolytic Capacitors

General Properties

- lower capacitance C at higher frequencies
- higher loss factors $\tan(\delta)$ than solid state capacitors
exception: „low ESR capacitors“
- leakage current falls with time
- „low leakage capacitors“ available



Example of a low leakage capacitor:

Vishay BCcomponents 013RLC - 100 μF /10V

$$\tan(\delta) = R_{\text{ESR}} \cdot \alpha \cdot \omega C \Rightarrow R_{\text{ESR}} \approx 1.3 \, \Omega \text{ at } 200 \, \text{Hz}$$

$$Z = R_{\text{ESR}} + \frac{1}{j\omega C} \Rightarrow R_{\text{ESR}} \approx 1.7 \, \Omega \text{ at } 10 \, \text{kHz}$$

ELECTRICAL DATA						
U_R (V)	C_R 100 Hz (μF)	NOMINAL CASE SIZE $\varnothing D \times L$ (mm)	I_R 100 Hz 85 °C (mA)	I_{L2} 2 min (μA)	$\tan \delta$ 100 Hz	Z 10 kHz (Ω)
10	47	5 x 11	75	1.0	0.16	2.8
	68	5 x 11	90	1.4	0.16	2.5
	100	5 x 11	110	2.0	0.16	1.7
	220	8.2 x 11	190	4.4	0.16	0.9

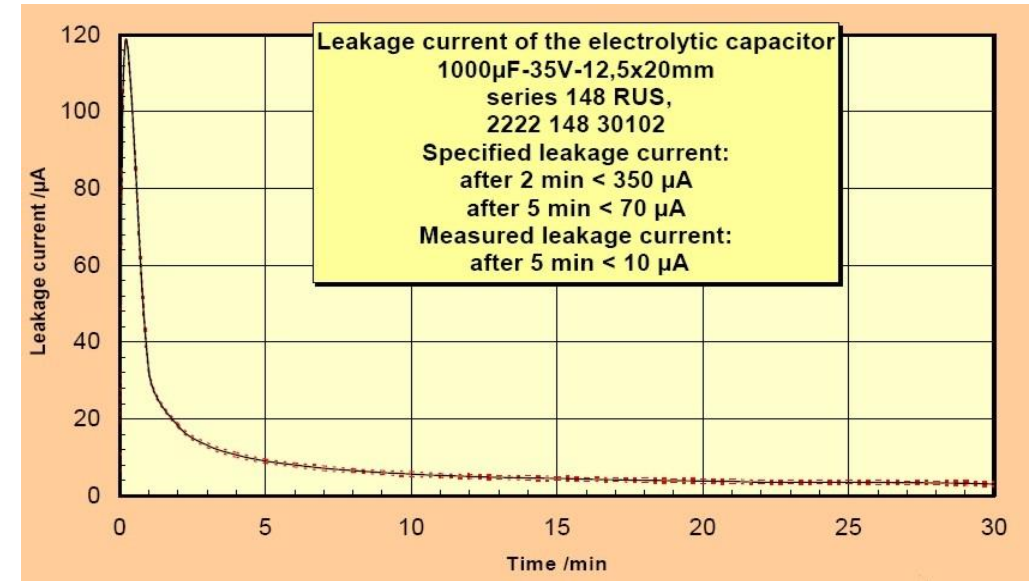
Electrolytic Capacitors

Typical Temporal Behaviour of the Leakage Current

- *Test conditions: current measurement after the application of the rated voltage*
- Analytic models for the maximum leakage current (provided by the supplier):

$$I_L(t = 2 \text{ min}) [\mu A] < (0.01 \cdot U_R[V] \cdot C_R[\mu F]) + 3 \mu A$$
$$I_L(t = 5 \text{ min}) [\mu A] < (0.002 \cdot U_R[V] \cdot C_R[\mu F]) + 3 \mu A$$

U_R : rated voltage
 C_R : rated capacitance



© Tadiran batteries

$$\Rightarrow \text{here: } I_L(t = 2 \text{ min}) = 353 \mu A$$
$$I_L(t = 5 \text{ min}) = 73 \mu A$$

$$\text{measured: } I_L(t = 5 \text{ min}) < 10 \mu A$$

Electrolytic Capacitors

Influences on the Leakage Current

Low leakage electrolytic capacitors

Example: Vishay BCcomponents 013 RLC

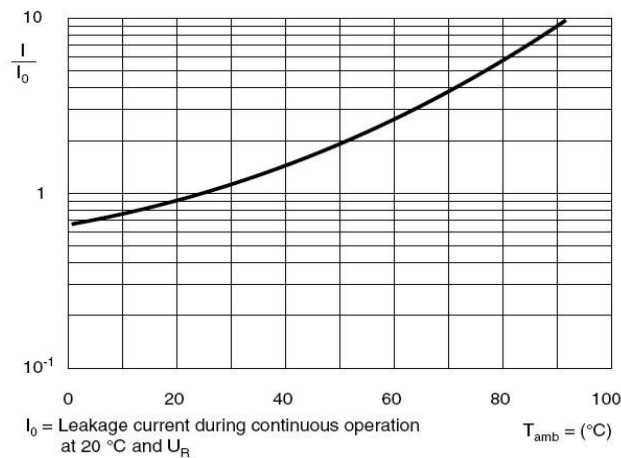
$$I_0(2 \text{ min}) [\mu A] < (0.002 \cdot U_R[V] \cdot C_R[\mu F]) \text{ or } 0.7 \mu A$$

whichever is greater

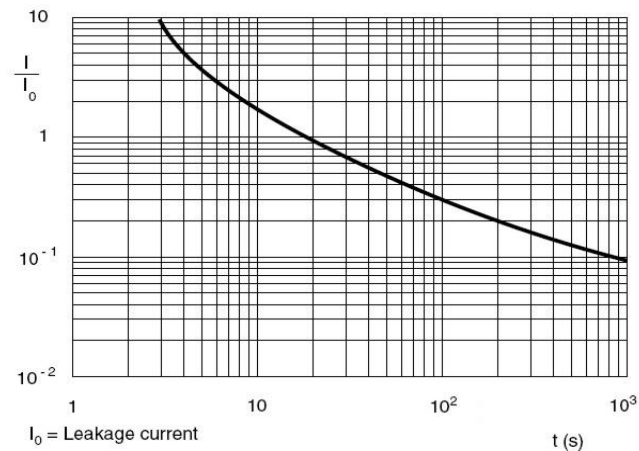
U_R : rated voltage (6.3...25 V)

C_R : rated capacitance

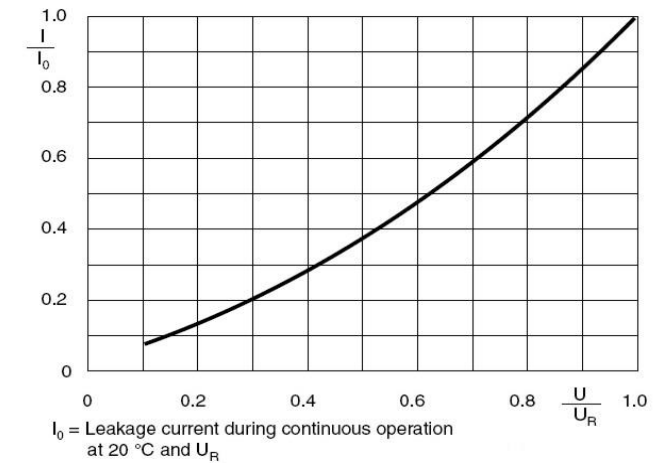
I_L as function of temperature



I_L as function of time



I_L as function of U_C



Electrolytic Capacitors

Influences on the Leakage Current

General behaviour of the leakage current

- high initial value after the first application of an electric voltage
- rapid drop within minutes to hours
- low continuous leakage current

Further influences

- Leakage current increases with temperature
- Leakage current increases with applied voltage

Consequences for the selection and treatment of electrolytic capacitors

- use of capacitors with higher rated voltage
➡ *lower applied voltage = lower leakage current*
- use of capacitors with higher values
➡ *lower charge-discharge rates = more stable, low leakage current*
- use of „burn-in“ (e.g. 1 hr. at rated voltage)
➡ *elimination of short-term effects*
- use of special „low leakage“ capacitors
➡ *optimized for lower continuous leakage currents*

Electrolytic Capacitors

Loss Calculation

Example: Vishay BCcomponents 013 RLC - 100 μ F/10V

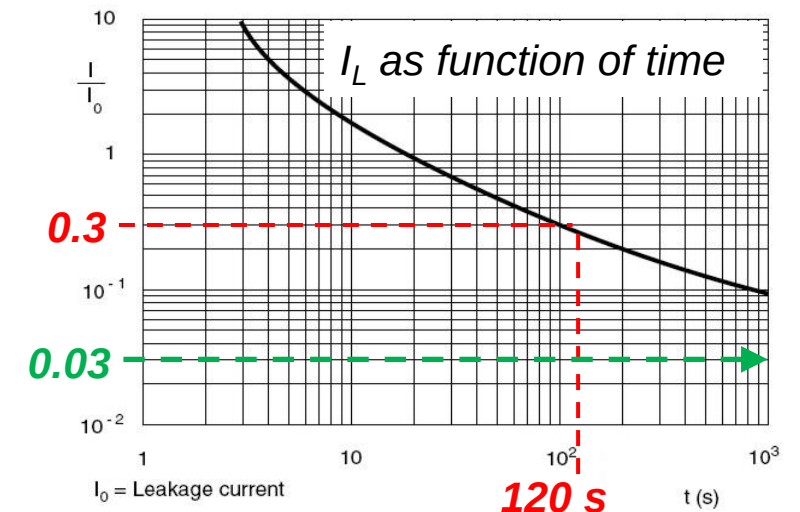
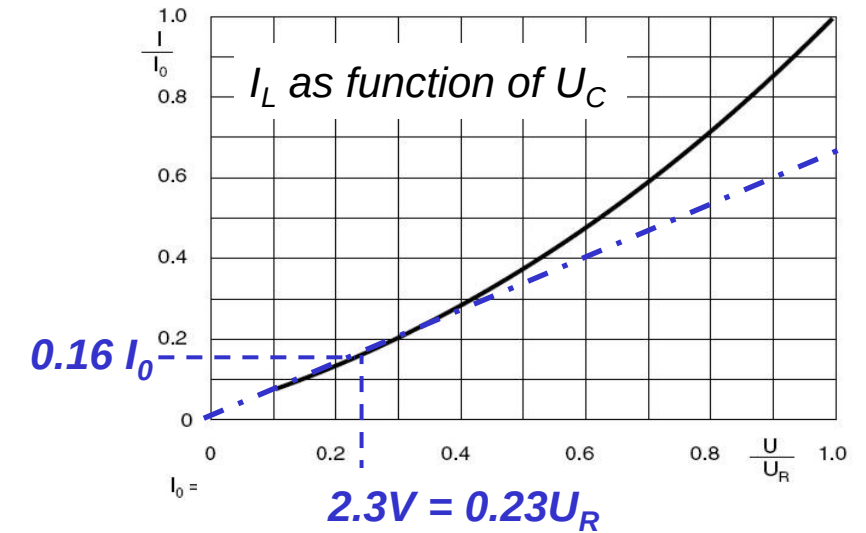
- buffered system voltage: 2.3 V
- active system power: 90 μ W
- standby power: 5 μ W

Power loss during operation:

after 2 minutes: $I_0 = 0,002 \cdot U_R \cdot C_R = 2 \mu\text{A}$
 assumption: $I_L(t \rightarrow \infty) = \frac{0,03}{0,3} \cdot 0,16 \cdot I_0 = 32 \text{ nA}$
 $U_C = 2.3 \text{ V} \Rightarrow \begin{cases} P_{\text{loss}} = U_C \cdot I_L = 74 \text{ nW} \\ R_p = U_C / I_L = 72 \text{ M}\Omega \end{cases}$

Time constants for self-discharge:

$\tau = R_p \cdot C = 72 \text{ M}\Omega \cdot 100 \mu\text{F} = 7200 \text{ s}; t_{50} = 0.35 \cdot \tau = 2520 \text{ s}$



Super Capacitors or „Gold caps“ Design

© Panasonic



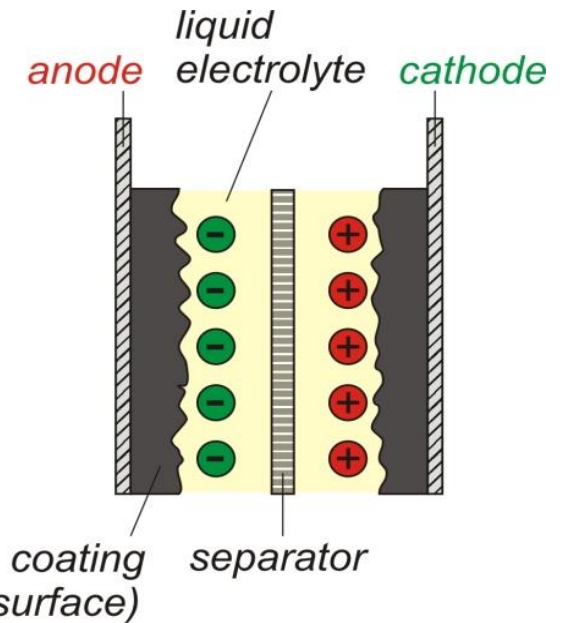
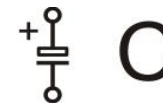
20 mm



© Cap-xx

Properties

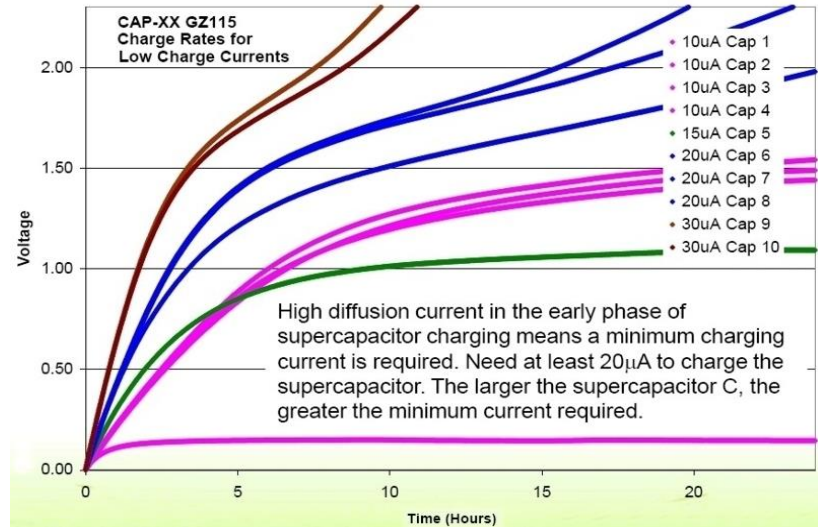
- use of Helmholtz double layers on high-surface electrodes (carbon, metal oxides or conductive polymer)
- liquid, organic or polymer electrolyte
- small double layer thickness: ~ a few nm
- large electrode area: ~ 2000 m²/g
- ➔ extremely high capacitance: ~ 0.01 ... 10 F
- relatively high leakage currents: ~ 0.1... 10 μA
- long (full) charging times: ~ several days



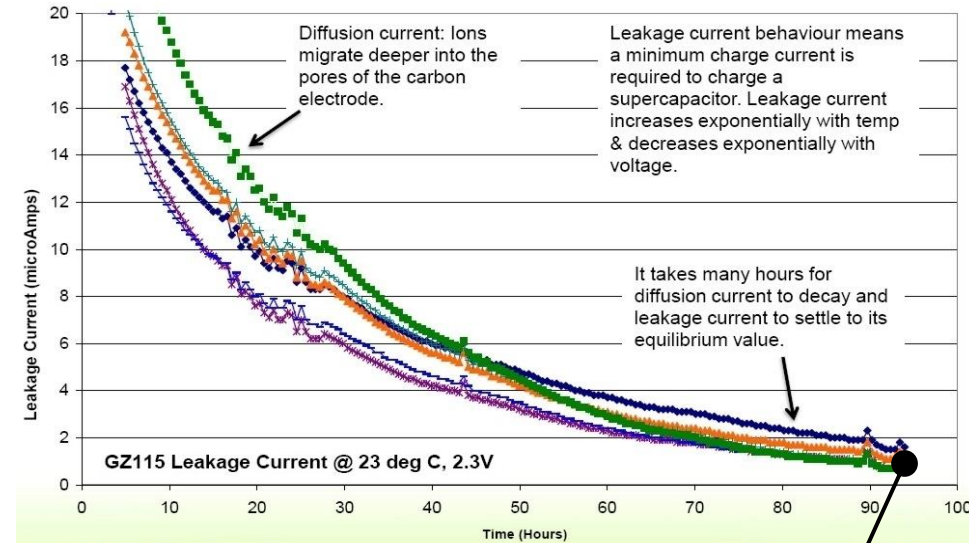
© Cap-xx

Super Capacitors

Leakage Current over Time



voltage vs. charging current

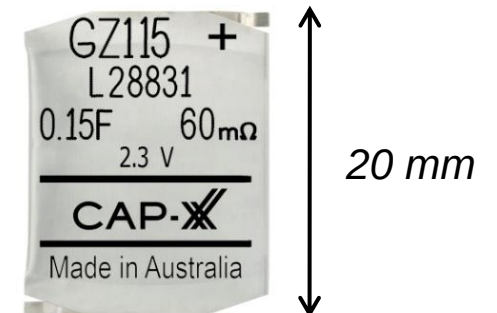


leakage current vs. time

1 μ A @ 4 days

Properties

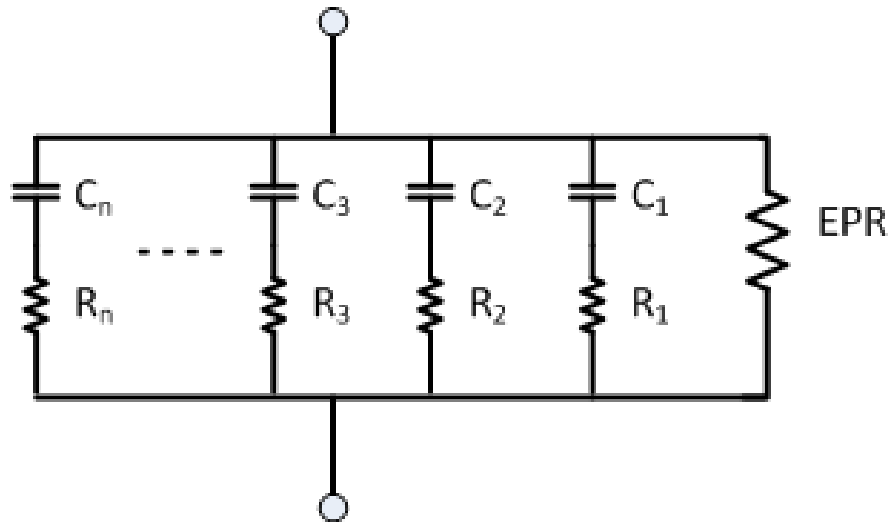
- minimum current required for charging: $> 10 \mu\text{A}$
- large leakage current over long time: $\sim 10 \dots 0.2 \mu\text{A}$
- long (full) charging time: $\sim$ several days



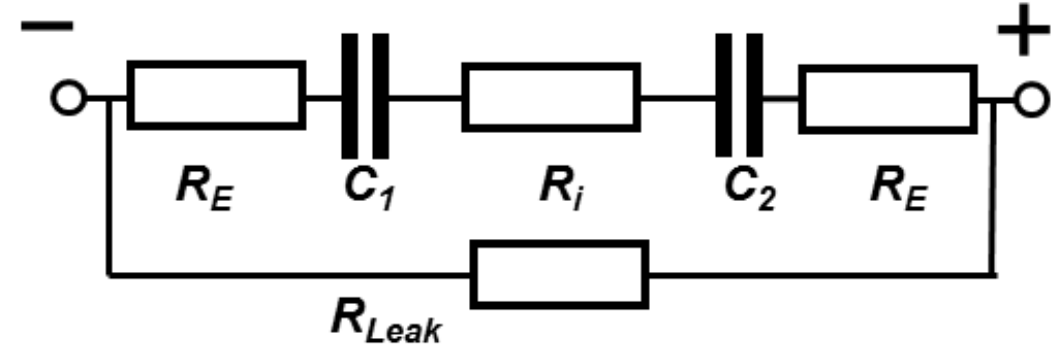
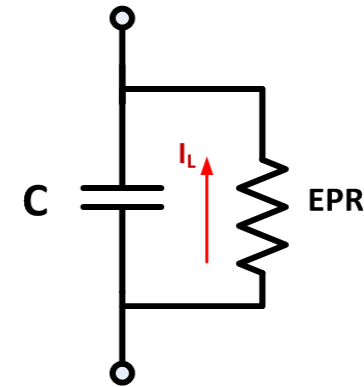
Super Capacitors

Equivalent Circuit

- Complex equivalent circuit
- **BUT** for slow discharge (order of seconds)
 - ➔ Simple equivalent circuit



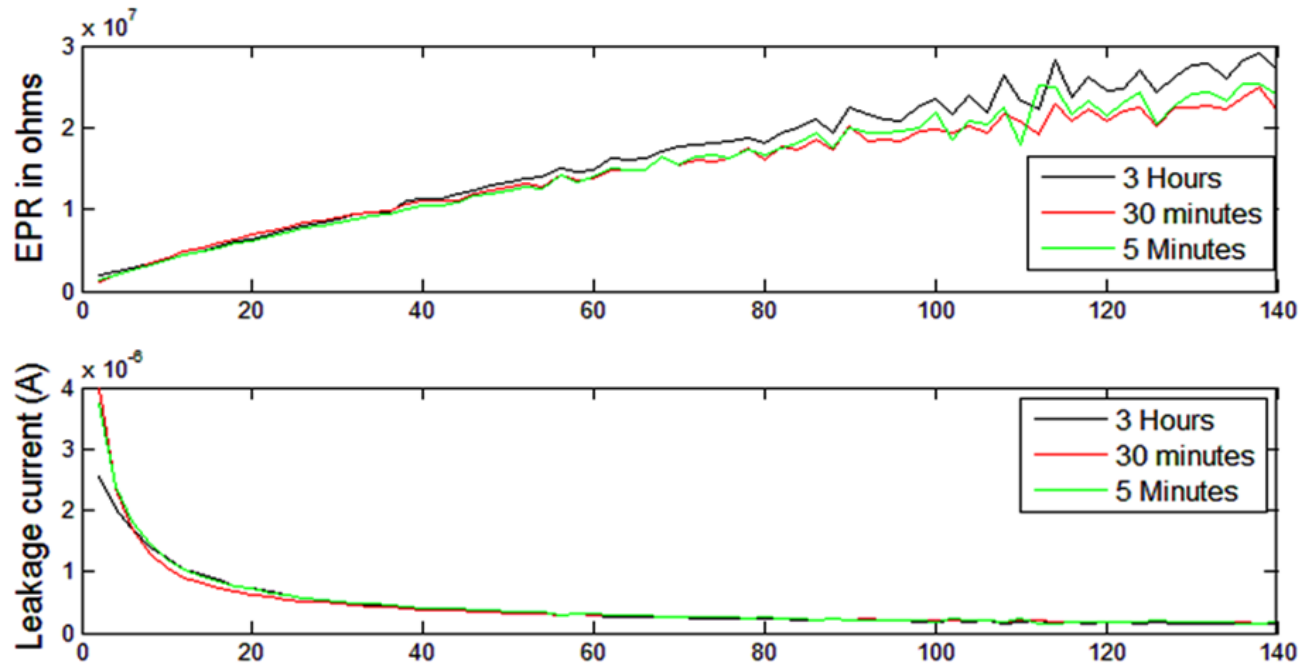
➔
(Spyker et al.)



Super Capacitors

Measurement

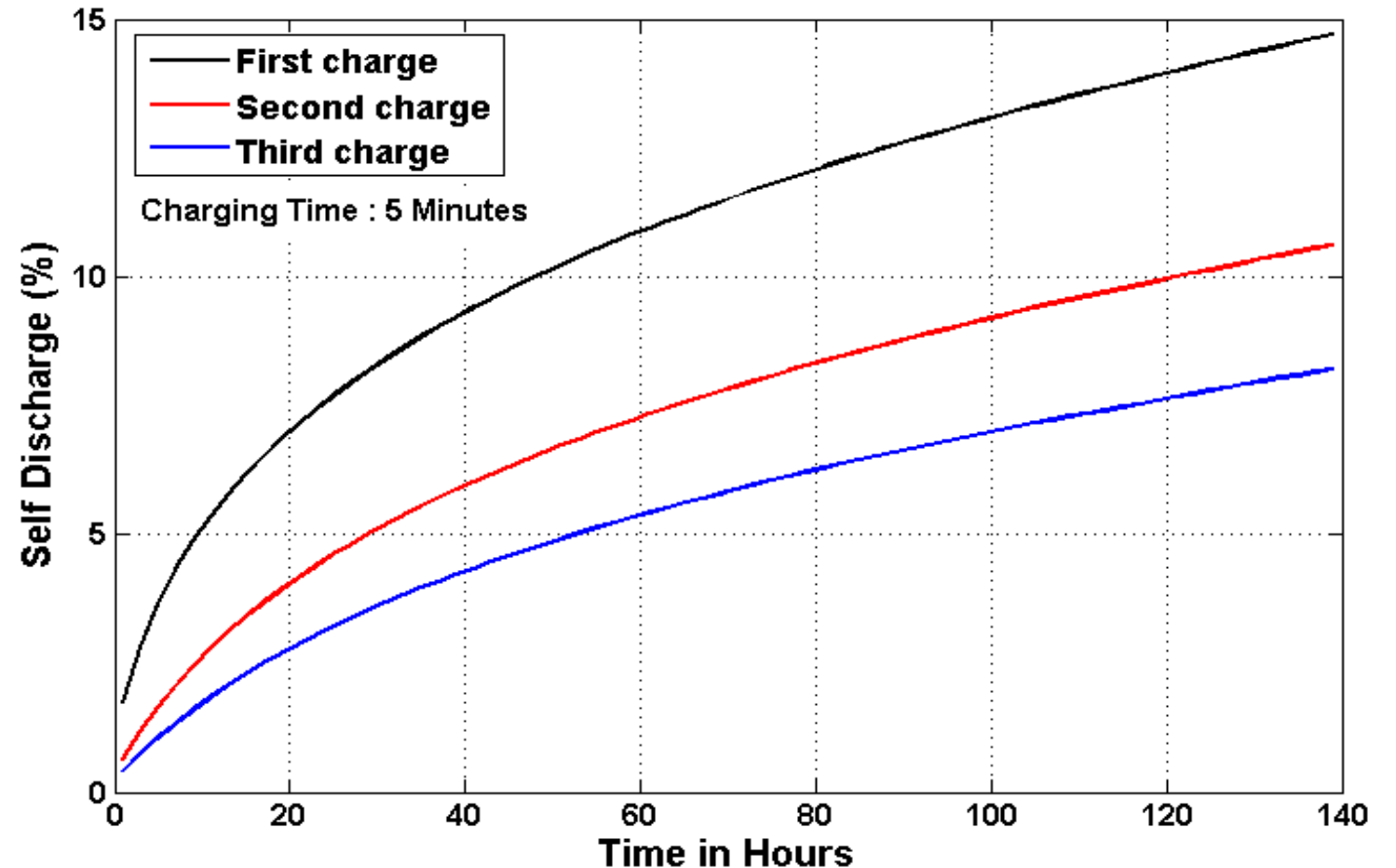
- EPR, I_L : Progressive (for different charging times)
- Successive charging $\rightarrow$ Self discharge
- Measurement condition: Constant temperature 21°C ($\pm 0.75^\circ \text{C}$)



Super Capacitors

Measurement

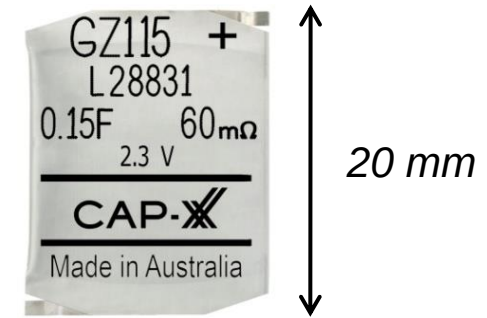
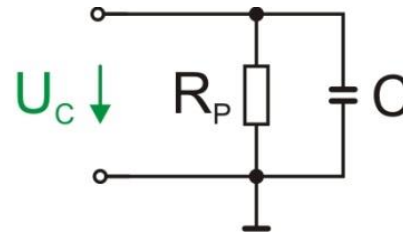
- Successive charging:
Conditioning of the Electrolytic
Double Layer (EDLC)



Super Capacitors

Long-term Loss Calculation

Example: Cap-xx GZ 115 - 0.15 F/2.3V



- from the supplier: best case leakage current $\sim 0.2 \mu\text{A}$ after 10 days: $R_p \approx \frac{2.3 \text{ V}}{0.2 \mu\text{A}} = 11.5 \text{ M}\Omega$
- power loss during long-term operation: $U_C = 2.3 \text{ V} \Rightarrow P_{loss} = \frac{U_C^2}{R_p} = 0.46 \mu\text{W}$
- discharge time constant: $\tau = R_p \cdot C = 11.5 \text{ M}\Omega \cdot 0.15 \text{ F} = 479 \text{ hrs}$
- 50% energy drop time: $t_{50} = 0.35 \cdot \tau = 168 \text{ hrs}$; $W_{loss} = \frac{1}{2} \cdot W_0 = 198 \text{ mJ}$

Capacitors

Conclusion

Choice of energy storage

- Long term leakage current can be very small for electrolytic capacitors
- Use pre-charged capacitors & keep them charged at a high energy level as good as possible
- Depends on the generator system's properties (U/I)
- Some need higher charging currents than others and/ or different voltages
- Leakage current might exceed charging current ☹
- Dimension your capacitors according to the energy needed for specific tasks (measure & send)

Next: What about batteries ?

- Thin-Film ?
- Leakage currents? Charging currents? Long-term stability? Energy density? Size?

